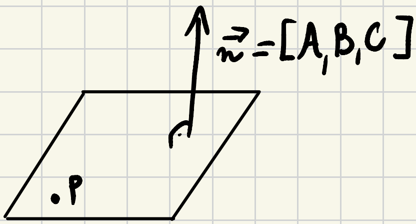
Płaszczyzna w R^3

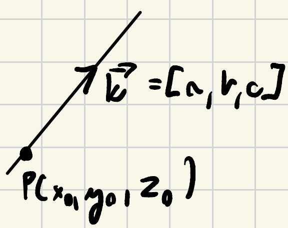
$P = (x_0, y_0, z_0)$



$\Pi: A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

$\Pi: Ax + By + Cz + D = 0$

Prosta



→ p. parametryczna

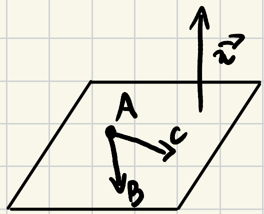
$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases} \quad t \in R$$

→ p. kierunkowa

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

→ p. krawędziowa

$$\vec{k} = \begin{vmatrix} i & j & k \\ A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \end{vmatrix} \quad \begin{cases} A_1 x + B_2 y + C_1 z + D_1 = 0 \\ A_2 x + B_2 y + C_2 z + D_2 = 0 \end{cases}$$



$\vec{n} \perp \vec{AB}$

$\vec{n} \perp \vec{AC}$

$\Downarrow$

$\vec{n} = \vec{AB} \times \vec{AC}$

Zad. 6

$A(0,1,0) \quad B(3,3,5) \quad C(1,3,-1)$

$\vec{AB} = [3,2,5]$

$\vec{AC} = [1,2,-1]$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 3 & 2 & 5 \\ 1 & 2 & -1 \end{vmatrix} \begin{matrix} i & j & k \\ 3 & 2 & 5 \end{matrix} = -2i + 6k + 5j - 2k - 10i + 3j =$$

$$= -12i + 8j + 4k$$

$$= \begin{bmatrix} -12 \\ 8 \\ 4 \end{bmatrix}$$

$$-12(x-0) + 8(y-1) + 4z = 0$$

$$-12x + 8y + 4z - 8 = 0$$

Zad. 7

$A(1,-1,1)$

$H_1: x-y+z=1$

$H_2: 2x+y+z=-1$

$\left.\begin{array}{l} H \perp H_1 \Rightarrow n \perp n_1 \\ H \perp H_2 \Rightarrow n \perp n_2 \end{array}\right| \Rightarrow n = n_1 \times n_2$

$n_1 = [1,-1,1]$

$n_2 = [2,1,1]$

$$\vec{n} = \begin{vmatrix} i & j & k \\ 1 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -i + k + 2j + 2k - i - j =$$

$$\begin{array}{ccc} i & j & k \\ 1 & -1 & 1 \end{array}$$

$$= -2i + j + 3k$$

$$\vec{n} = [-2,1,3]$$

$$-2(x-1)+1(y+1)+3(z-1)=0$$

$$-2x+2+y+1+3z-3=0$$

$$-2x+y+3z=0$$

Wzajemne położenie prostych l_1 i l_2

- $l_1 \parallel l_2 \iff \vec{k_1} = \vec{k_2} \iff \exists\, m \in \mathbb{R} \quad \vec{k_1} = m \cdot \vec{k_2} \quad \left(\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}\right)$
- $l_1 \perp l_2 \iff \vec{k_1} \perp \vec{k_2} \iff \vec{k_1} \circ \vec{k_2} = 0$
- proste się przecinają pod $\sphericalangle \varphi \iff \exists$ pkt. wspólny
- wpw $\rightarrow$ proste są skośne

Zad. 2

$$l_1: \; -(x+1) = y-1 = \frac{z-1}{2} \qquad\qquad l_2: \; \frac{x}{2} = y = -z$$

$$\vec{k_1} = [-1, 1, 2] \qquad\qquad \vec{k_2} = [2, 1, -1]$$

$$\vec{k_1} = m \cdot \vec{k_2} \text{ dla } m \in \emptyset \qquad l_1 \nparallel l_2$$

$$\vec{k_1} \circ \vec{k_2} = -2 + 1 - 2 \neq 0 \qquad l_1 \not\perp l_2$$

$$l_1 \begin{cases} x = -1 - t \\ y = 1 + t \\ z = 1 + 2t \end{cases} \qquad\qquad l_2 \begin{cases} x = 2t \\ y = t \\ z = -t \end{cases}$$

l_1 i l_2 są skośne
$$\begin{cases} -1 - t = 2t \implies t = -\frac{1}{3} \\ 1 + t = t \implies 1 \neq 0 \\ 1 + 2t = -t \implies t = -\frac{1}{3} \end{cases}$$

Układ sprzeczny

Zad. 3

$(0,-1,-1)$

$H_1: x+y-z=0$

$H_2: x+y=-1$

$\begin{cases} x+2y-3z=0 \\ 2x+y=0 \end{cases}$ $(0,0,0)$

$H_3: x+2y-3z=0$

$2x+y=0$

$$\vec{k_1} = \begin{vmatrix} i & j & k \\ 1 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix} = k - j - k + i = i - j + 0k$$

$$\begin{matrix} i & j & k \\ 1 & 1 & -1 \end{matrix}$$

$\vec{k_1} = [1,-1,0]$

$$\vec{k_2} = \begin{vmatrix} i & j & k \\ 1 & 2 & -3 \\ 2 & 1 & 0 \end{vmatrix} = k - 6j - 4k + 3i = 3i - 6j - 3k$$

$$\begin{matrix} i & j & k \\ 1 & 2 & -3 \end{matrix}$$

$\vec{k_2} = [3,-6,-3]$

$\vec{k_1} = m \cdot \vec{k_2}$ dla $m \in \emptyset$ $k_1 \nparallel k_2$

$\vec{k_1} \circ \vec{k_2} = 3 + 6 + 0 = 9 \neq 0$ $k_1 \not\perp k_2$

$$l_1: \begin{cases} x = t \\ y = -1-t \\ z = -1 \end{cases} \qquad l_2: \begin{cases} x = 3t \\ y = -6t \\ z = -3t \end{cases}$$

$$\begin{cases} t = 3t & \Rightarrow t = 0 \\ -1-t = -6t & \Rightarrow t = \frac{1}{5} \\ -1 = -3t & \Rightarrow t = \frac{1}{3} \end{cases}$$

Układ nie ma rozwiązań, więc brak punktów wspólnych

l_1 i l_2 są skośne

22 stycznia
2 kartki A3
kalkulator

Zad. 5 $P(3, 0, -1)$

$H_1: x + y - 2z = 4 \Rightarrow n_1 = [1, 1, -2]$

$H_2: x + 4y + 3z = 3 \Rightarrow n_2 = [1, 4, 3]$

$$n = n_1 \times n_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & -2 \\ 1 & 4 & 3 \\ i & j & k \\ 1 & 1 & -2 \end{vmatrix} = 3i + 4k - 2j - k + 8i - 3j =$$

$$= 11i - 5j + 3k$$

$n = [11, -5, 3]$

$$\begin{cases} x = 3 + 11t \\ y = -5t \\ z = -1 + 3t \end{cases}$$

Zad. 6

$l: \frac{x-1}{4} = \frac{y+3}{-3} = \frac{z}{2}$ $\qquad$ $H: 2x+4y+2z=5$

$\vec{k} = [4,-3,2]$ $\qquad$ $\vec{n} = [2,4,2]$

$\vec{k} = m \cdot \vec{n}$ dla $m \in \emptyset$ $\quad \vec{k} \not\parallel \vec{n}$

$\vec{k} \circ \vec{n} = 8 - 12 + 4 = 0$ $\quad \vec{k} \perp \vec{n}$, więc $l \parallel H$

!

$$\boxed{\begin{matrix} H \parallel l \Leftrightarrow \vec{n} \perp \vec{k} \\ H \perp l \Leftrightarrow \vec{n} \parallel \vec{k} \end{matrix}}$$

Zbadaj wzajemne położenie prostych k. i l oraz jeśli to możliwe równanie płaszczyzny, która zawiera

$k: \frac{x+2}{3} = \frac{y}{2} = z-1$ $\qquad$ $\vec{k_1} = [3,2,1]$

$l: \begin{cases} x = 1+t \\ y = 2 \\ z = 2-t \end{cases}$ $\qquad$ $\vec{k_2} = [1,0,-1]$

$\vec{k_1} = m \cdot \vec{k_2}$ dla $m \in \emptyset$ $\vec{k_1} \nparallel \vec{k_2}$

$\vec{k_1} \circ \vec{k_2} = 3 - 1 = 2 \neq 0$ $\vec{k_1} \not\perp \vec{k_2}$

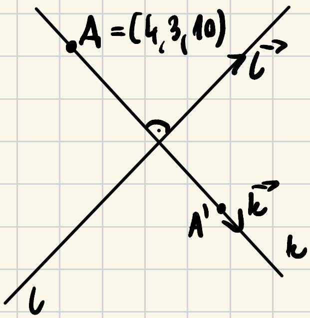
$$k: \begin{cases} x = -2 + 3t \\ y = 2t \\ z = 1 + t \end{cases} \qquad l: \begin{cases} x = 1 + t \\ y = 2 \\ z = 2 - t \end{cases}$$

$$\begin{cases} -2 + 3t = 1 + t & \Rightarrow 2t = 3 \\ 2t = 2 & \Rightarrow t = 1 \\ 1 + t = 2 - t & \Rightarrow 2t = 1 \end{cases}$$

Brak rozwiązania, nie mają punktów wspólnych

$\Downarrow$

H nie istnieje



TEGO NIE MA

NA KOLOSIE

$\vec{l} = [2, 4, 5]$

$\vec{k} \perp \vec{l} \iff 2a + 4b + 5c = 0$

$P_0(0, 0, \frac{1}{2})$

$$l: \begin{cases} x = 2t \\ y = 4t \\ z = \frac{1}{2} + 5t \end{cases} \qquad k: \begin{cases} x = 4 + at \\ y = 3 + bt \\ z = 10 + ct \end{cases}$$

$$\begin{cases} 4 + at = 2t \\ 3 + bt = 4t \\ 10 + ct = \frac{1}{2} + 5t \\ 2a + 4b + 5a = 0 \end{cases}$$